



ML & DL

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635

Lab - 8

Step 1: Import Libraries

This step imports all necessary libraries for data processing, visualization, and machine learning.

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree
```

Step 2: Load the Dataset

Load Given dataset - heart.csv

```
In [2]: df = pd.read_csv("heart.csv")
df
```

Out[2]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tha
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	59	1	1	140	221	0	1	164	1	0.0	2	0	2
1021	60	1	0	125	258	0	0	141	1	2.8	1	1	3
1022	47	1	0	110	275	0	0	118	1	1.0	1	1	2
1023	50	0	0	110	254	0	0	159	0	0.0	2	0	2
1024	54	1	0	120	188	0	1	113	0	1.4	1	1	3

1025 rows × 14 columns



Step 3: Data Overview

In this step, we examine the dataset structure, summary statistics, and check for missing values.

In [3]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   age         1025 non-null   int64
 1   sex         1025 non-null   int64
 2   cp          1025 non-null   int64
 3   trestbps    1025 non-null   int64
 4   chol        1025 non-null   int64
 5   fbs         1025 non-null   int64
 6   restecg     1025 non-null   int64
 7   thalach     1025 non-null   int64
 8   exang       1025 non-null   int64
 9   oldpeak     1025 non-null   float64
10   slope       1025 non-null   int64
11   ca          1025 non-null   int64
12   thal        1025 non-null   int64
13   target      1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

In [4]: `df.describe()`

Out[4]:

	age	sex	cp	trestbps	chol	fbs	restecg
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	54.434146	0.695610	0.942439	131.611707	246.000000	0.149268	0.500000
std	9.072290	0.460373	1.029641	17.516718	51.59251	0.356527	0.500000
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	48.000000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	56.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	275.000000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

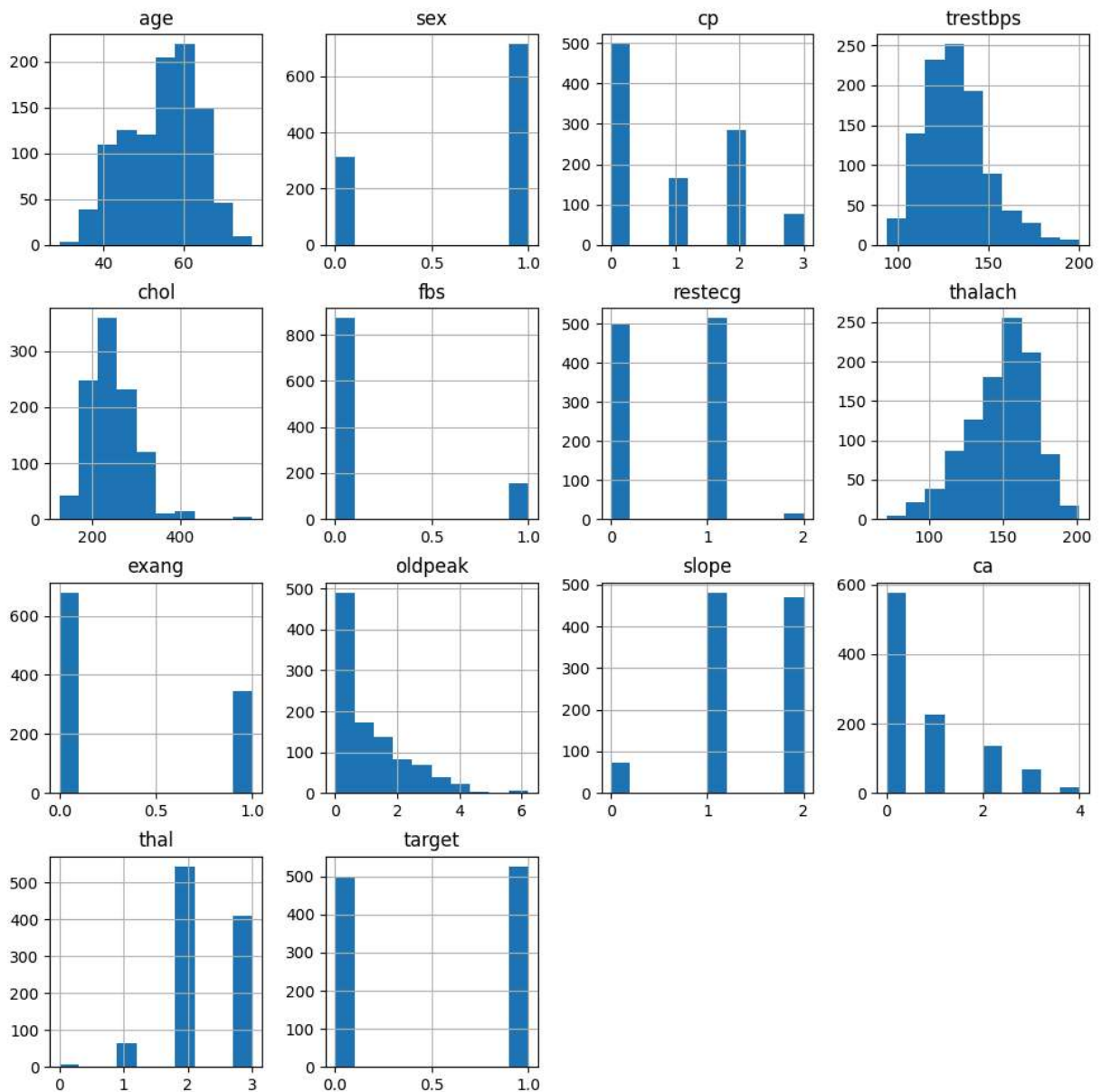
In []:

Step 4: Univariate Analysis

Here we visualize the distribution of each feature using histograms.

In [5]: `df.hist(figsize = (12,12))`

```
Out[5]: array([[<Axes: title={'center': 'age'}>, <Axes: title={'center': 'sex'}>,
<Axes: title={'center': 'cp'}>,
<Axes: title={'center': 'trestbps'}>],
[<Axes: title={'center': 'chol'}>,
<Axes: title={'center': 'fbs'}>,
<Axes: title={'center': 'restecg'}>,
<Axes: title={'center': 'thalach'}>],
[<Axes: title={'center': 'exang'}>,
<Axes: title={'center': 'oldpeak'}>,
<Axes: title={'center': 'slope'}>,
<Axes: title={'center': 'ca'}>],
[<Axes: title={'center': 'thal'}>,
<Axes: title={'center': 'target'}>, <Axes: >, <Axes: >]],
dtype=object)
```



Step 5: Bivariate Analysis

This step involves exploring the correlations between features using a heatmap.

```
In [6]: cols = df.columns
```

```
In [7]: %matplotlib qt
```

```
In [8]: sns.heatmap(df.corr(),annot=True)
```

```
Out[8]: <Axes: >
```

Step 6: Outlier Detection

We visualize potential outliers using boxplots.

```
In [9]: sns.boxplot(df)
```

```
Out[9]: <Axes: >
```

Step 7: Split Data into Training and Testing Sets

The dataset is split into training and testing sets for model evaluation.

```
In [10]: x = df.iloc[:,0:-1]
```

```
In [11]: y = df.iloc[:, -1]
```

```
In [12]: from sklearn.model_selection import train_test_split  
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=42)
```

Step 8: Train Decision Tree

We train a Decision Tree Classifier on the training data. You have to also check for KNeighborsClassifier and GaussianNB

```
In [13]: from sklearn.tree import DecisionTreeClassifier
```

```
In [14]: df = DecisionTreeClassifier()
```

```
In [15]: df.fit(x_train, y_train)
```

Out[15]:

▼ DecisionTreeClassifier ⓘ ?

► Parameters

In [16]: `y_pred = df.predict(x_test)`In [17]: `from sklearn.metrics import accuracy_score`In [18]: `print("accuracy : ", accuracy_score(y_test, y_pred))`

accuracy : 0.9853658536585366

In [19]: `df.score(x_test, y_pred)`

Out[19]: 1.0

Cross Validation

In [20]: `from sklearn.model_selection import KFold`In [21]: `k = KFold(n_splits=10)`In [22]: `x`

Out[22]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tha
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	59	1	1	140	221	0	1	164	1	0.0	2	0	2
1021	60	1	0	125	258	0	0	141	1	2.8	1	1	3
1022	47	1	0	110	275	0	0	118	1	1.0	1	1	2
1023	50	0	0	110	254	0	0	159	0	0.0	2	0	2
1024	54	1	0	120	188	0	1	113	0	1.4	1	1	3

1025 rows × 13 columns



```
In [23]: for train_index, test_index in k.split(x):
          x_train, x_test = x.values[train_index], x.values[test_index]
          y_train, y_test = y[train_index], y[test_index]
          model1 = DecisionTreeClassifier()
          model1.fit(x_train, y_train)
          print(model1.score(x_test, y_test))
```

```
1.0
1.0
1.0
1.0
1.0
1.0
1.0
1.0
1.0
1.0
```

```
In [24]: from sklearn.model_selection import cross_val_score
```

```
In [25]: cross_val_score(df, x, y, cv=10)
```

```
Out[25]: array([1., 1., 1., 1., 1., 1., 1., 1., 1., 1.])
```

```
In [ ]:
```

```
In [26]: x1 = np.arange(1,11).reshape(-1,1)
          x1
```

```
Out[26]: array([[ 1],
                 [ 2],
                 [ 3],
                 [ 4],
                 [ 5],
                 [ 6],
                 [ 7],
                 [ 8],
                 [ 9],
                 [10]])
```

```
In [27]: y1 = np.array([0,1,0,1,0,1,0,1,0,1])
          y1
```

```
Out[27]: array([0, 1, 0, 1, 0, 1, 0, 1, 0, 1])
```

```
In [28]: for i,j in k.split(x1):
          print("<----->")
          print(x1[i])
          print(x1[j])
          x_train, x_test = x1[i], x1[j]
          y_train, y_test = y1[i], y1[j]
```

```
<----->
[[ 2]
 [ 3]
 [ 4]
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 [ 8]
 [ 9]
 [10]]
[[1]]
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[[ 1]
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[[5]]
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```



```
[[ 1]
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 [8]
 [9]]
[[10]]
```

BS4 | web Scraping to get Data

```
In [29]: import requests  
import bs4
```

```
In [30]: req = requests.get("https://darshan.ac.in/faculty-list/btech-computer")
```

```
In [31]: req.status_code
```

```
Out[31]: 200
```

```
In [32]: soup = bs4.BeautifulSoup(req.text, "lxml")
```

```
In [33]: data = soup.select("body > main > div:nth-child(5) > div > div > div > div > div")
```

```
In [34]: len(data)
```

```
Out[34]: 31
```

```
In [35]: for row in data:  
    name = row.select("h2")[0].text.strip()  
    other = row.select("span")  
    print(name, "---->", other[0].text.strip())
```

Dr. Gopi Sanghani ----> Dean - Faculty of Computer Science
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Prof. SAMIR KARIYA ----> Assistant Professor

In []: